1. Primero construimos la población completa con `label_3m`

```
full_population_labeled <- sdf_sql(
  sc,
  "
  WITH eligible AS (

     /* ============================================================
        FULL ELIGIBLE POPULATION BEFORE 10:1 NEGATIVE SAMPLING
        ============================================================ */

     SELECT DISTINCT
        party_id,
        observation_date

     FROM <PRE_SAMPLING_ELIGIBLE_TABLE>

     WHERE observation_date BETWEEN '2022-01-01' AND '2025-12-31'
  ),

  labeled AS (

     SELECT
        e.party_id,
        e.observation_date,

        YEAR(e.observation_date)  AS observation_year,
        MONTH(e.observation_date) AS observation_month,

        f.first_qualifying_alert_date,

        CASE
           WHEN f.first_qualifying_alert_date > e.observation_date
            AND f.first_qualifying_alert_date
                <= ADD_MONTHS(e.observation_date, 3)
           THEN 1
           ELSE 0
        END AS label_3m

     FROM eligible e

     LEFT JOIN bu_frd.party_first_valid_alert_all f
        ON e.party_id = f.party_id
  )

  SELECT *
  FROM labeled
```

"
)

full_population_labeled

Aquí **no hacemos sampling**. Todos los elegibles permanecen en la población.

---

2. Tasa real por año

```
annual_outcome_rate <- full_population_labeled %>%
  group_by(observation_year) %>%
  summarise(
    eligible_population = n(),
    label_1 = sum(label_3m),
    label_0 = n() - sum(label_3m),
    outcome_rate = mean(label_3m),
    outcome_rate_pct = 100 * mean(label_3m)
  ) %>%
  arrange(observation_year) %>%
  collect()

annual_outcome_rate
```

El resultado debería verse así:

observation_year	eligible_population	label_1	label_0	outcome_rate_pct
2022	...	...	...	...
2023	...	...	...	...
2024	...	...	...	...
2025	...	...	...	...

Esta sí sería:

$$\boxed{ \text{Real Annual Outcome Rate} = \frac{Label1}{Full\ Eligible\ Population} }$$

---

3. Tasa real por `observation_date` mensual

Yo recomiendo mantener la fecha completa para no mezclar enero de distintos años.

```
monthly_outcome_rate <- full_population_labeled %>%
  group_by(
    observation_year,
```

```
  observation_month,
  observation_date
) %>%
summarise(
  eligible_population = n(),
  label_1 = sum(label_3m),
  label_0 = n() - sum(label_3m),
  outcome_rate = mean(label_3m),
  outcome_rate_pct = 100 * mean(label_3m)
) %>%
arrange(observation_date) %>%
collect()

monthly_outcome_rate
```

Esperamos aproximadamente **48 filas**:

Jan-2022
Feb-2022
...
Dec-2025

---

4. Yo correría este QA inmediatamente después

Queremos confirmar que los positives de la población completa coinciden con los positives que ya tenemos en la base final.

```
positive_qa <- sdf_sql(
  sc,
  "
  WITH full_positive AS (

    SELECT
      YEAR(observation_date) AS observation_year,
      COUNT(*) AS full_population_positives

    FROM (
      SELECT
        e.party_id,
        e.observation_date,

        CASE
          WHEN f.first_qualifying_alert_date > e.observation_date
           AND f.first_qualifying_alert_date
               <= ADD_MONTHS(e.observation_date, 3)
          THEN 1
          ELSE 0
```

```
            END AS label_3m

        FROM (
            SELECT DISTINCT
                party_id,
                observation_date

            FROM <PRE_SAMPLING_ELIGIBLE_TABLE>

            WHERE observation_date
                BETWEEN '2022-01-01' AND '2025-12-31'
        ) e

        LEFT JOIN bu_frd.party_first_valid_alert_all f
            ON e.party_id = f.party_id
    ) x

    WHERE label_3m = 1

    GROUP BY YEAR(observation_date)
),

final_positive AS (

    SELECT
        YEAR(observation_date) AS observation_year,
        COUNT(*) AS final_sample_positives

    FROM bu_frd.model_features_ab_final_2022_2025

    WHERE label_3m = 1

    GROUP BY YEAR(observation_date)
)

SELECT
    f.observation_year,

    f.full_population_positives,
    s.final_sample_positives,

    f.full_population_positives
        - s.final_sample_positives
        AS difference

FROM full_positive f

LEFT JOIN final_positive s
```

```
    ON f.observation_year = s.observation_year

  ORDER BY f.observation_year
  "
) %>%
  collect()

positive_qa
```

Idealmente debería producir:

Year	Full population positives	Final sample positives	Difference
2022	13,193	13,193	0
2023	13,840	13,840	0
2024	23,557	23,557	0
2025	12,009	12,009	0

Si la diferencia es `0`, confirmamos que el sampling únicamente redujo los negatives.

---

5. QA de número de observation months

También correría:

```
observation_month_qa <- full_population_labeled %>%
  summarise(
    total_rows = n(),
    distinct_observation_dates = n_distinct(observation_date),
    min_observation_date = min(observation_date),
    max_observation_date = max(observation_date)
  ) %>%
  collect()

observation_month_qa
```

Esperaría:

```
distinct_observation_dates = 48
min = 2022-01-31
max = 2025-12-31
```

---

6. Muy importante antes de interpretar 2025

También comprobaría que tenemos suficiente información futura de alerts para diciembre de 2025.

```
alert_history_qa <- sdf_sql(
  sc,
  "
  SELECT
      MIN(first_qualifying_alert_date) AS min_alert_date,
      MAX(first_qualifying_alert_date) AS max_alert_date

  FROM bu_frd.party_first_valid_alert_all
  "
) %>%
  collect()

alert_history_qa
```

Para clasificar correctamente:

observation_date = 2025-12-31

necesitamos poder observar hasta aproximadamente:

2026-03-31

Si la fuente cubre al menos ese período, 2025 completo puede evaluarse.

Qué me interesa que me pases primero

Una vez lo corras, pásame principalmente estos dos outputs:

1. `annual_outcome_rate`
2. `monthly_outcome_rate`

y especialmente los meses de **2023, 2024 y 2025**.

Con esos números podremos determinar si el pico que vimos en 2024 fue simplemente mayor volumen absoluto o si realmente aumentó la **tasa del outcome dentro de la población completa**.